SPH Bird Strike Simulation on a NACA 2412 Wing

Comparison with the leading edge impact case of Walvekar et al.

Kevin Mathew

Abstract

A bird impact on a simplified NACA 2412 wing shell was simulated with smoothed particle hydrodynamics (SPH) in Ansys Explicit Dynamics. The bird travelled at 213.66 m/s along the chordwise direction. At the final saved frame, the wing displacement reached 14.943 mm and the maximum displayed shell von Mises stress was 1,064.8 MPa. The closest case by speed in Walvekar et al. [1] used a 219.66 m/s oblique strike on a built-up leading edge. This report compares the impact inputs and structural response. Experimental validation of damage remains open because the models differ in target construction, impact angle and material behaviour.

Model and analysis setup

Item	Present model
Geometry and support	NACA 2412 section with 500 mm chord and 500 mm span. The 2 mm shell was fixed along both open spanwise ends. The bird started 10 mm from the leading edge.
Bird representation	Water-like SPH material at 950 kg/m³ with a linear shock equation of state. The flat-ended cylinder measured 113 mm in diameter and 190 mm in length.
Wing discretisation	5,018 nodes and 4,933 shell elements (4,909 quadrilaterals and 24 triangles), with three shell sublayers.
Bird discretisation	3,576 particles at the 8 mm particle-size setting. Geometric mass was approximately 1.81 kg and solver mass was 1.739 kg.
Material and contact	Linear elastic aluminium and frictionless trajectory contact. Geometric-strain erosion was enabled at 1.5. Material failure erosion was disabled.
Solution	Ansys Explicit Dynamics 2026 R1 Student (Autodyn). Mass scaling was off. The final frame was saved at 3.0004 ms, cycle 3597.

Displacement and particle dispersion

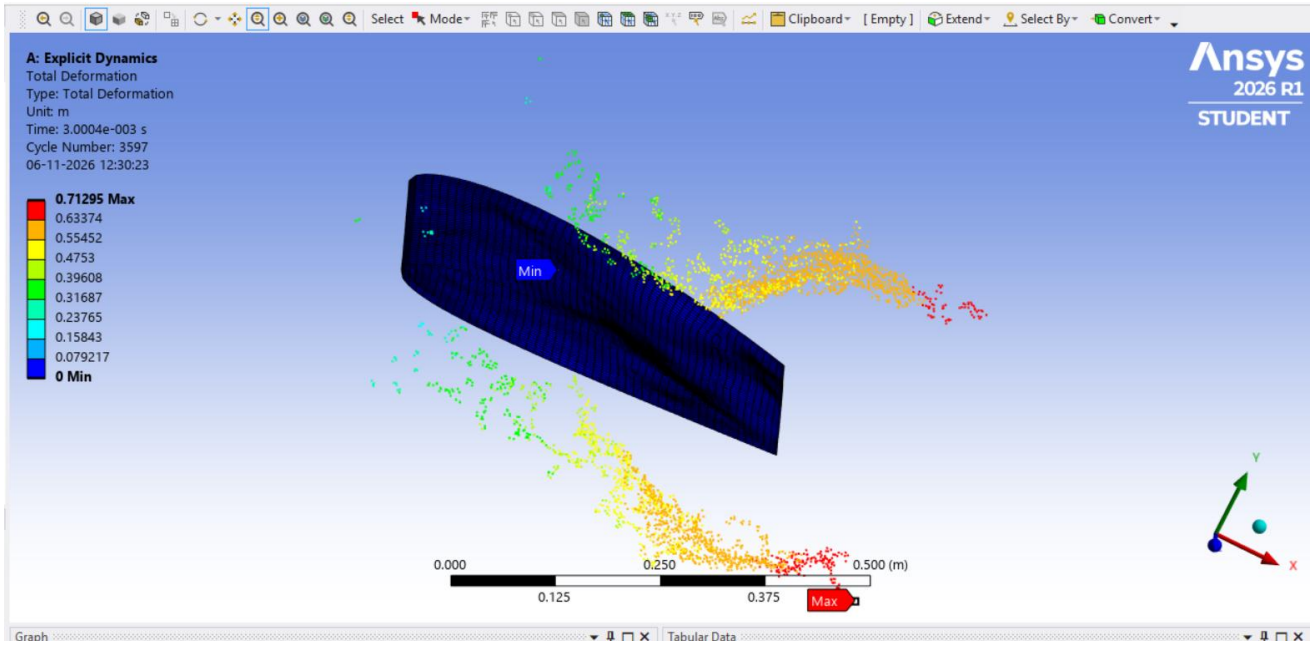


Figure 1. Total displacement for the bird and wing at 3.0004 ms. The 712.95 mm maximum is on a bird particle.

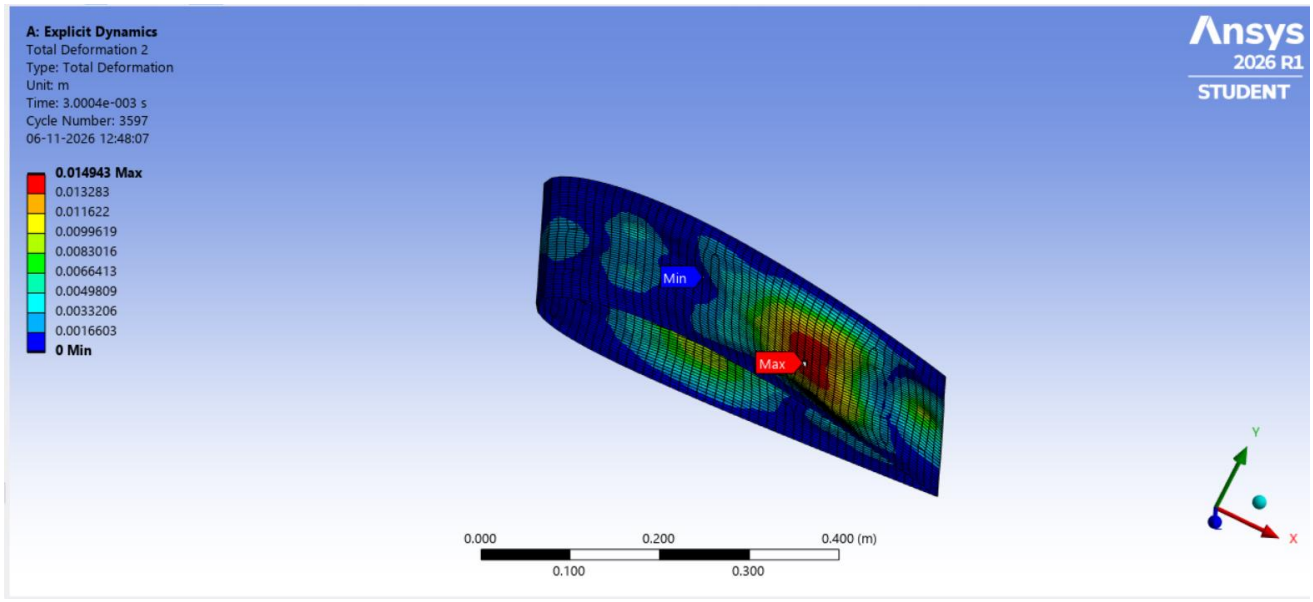


Figure 2. Wing total displacement at 3.0004 ms. The maximum displacement magnitude is 14.943 mm.

The full-model contour shows particle dispersion around the shell. The wing-only contour uses its own displacement scale. The reported shell displacement is a spatial maximum at the saved frame. A residual dent measurement would require a defined location and direction after unloading.

Stress and plastic strain

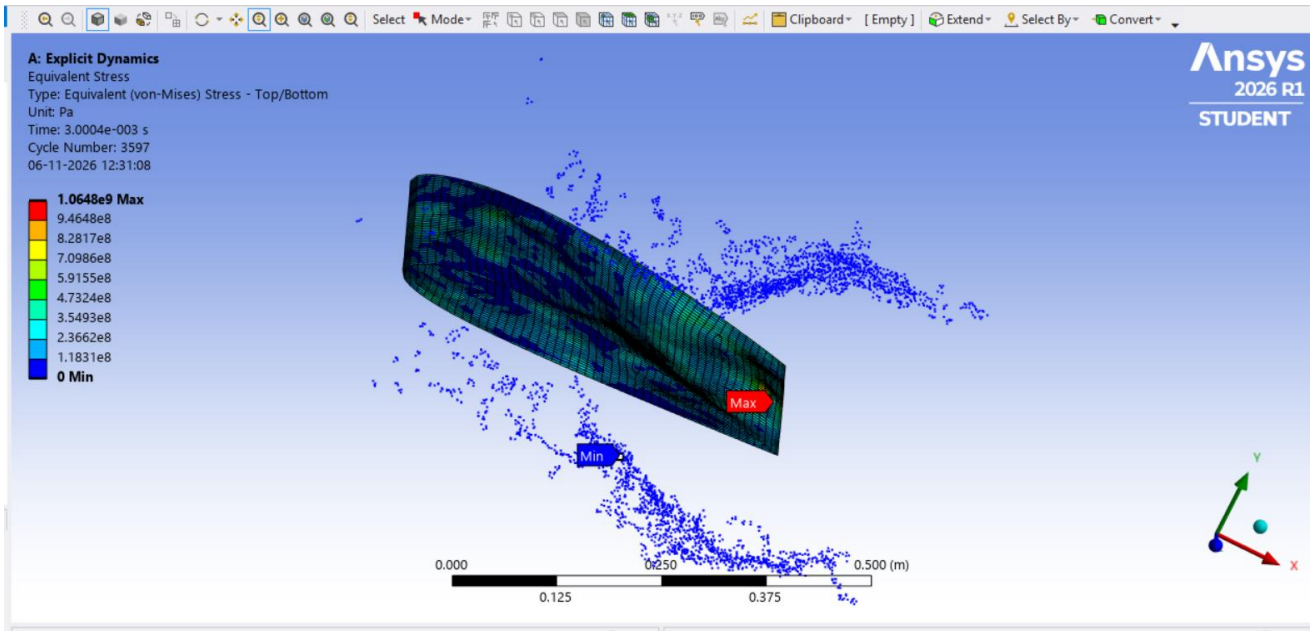


Figure 3. Equivalent von Mises stress at 3.0004 ms. The maximum marker lies on the shell. Bird particles remain visible in this plot.

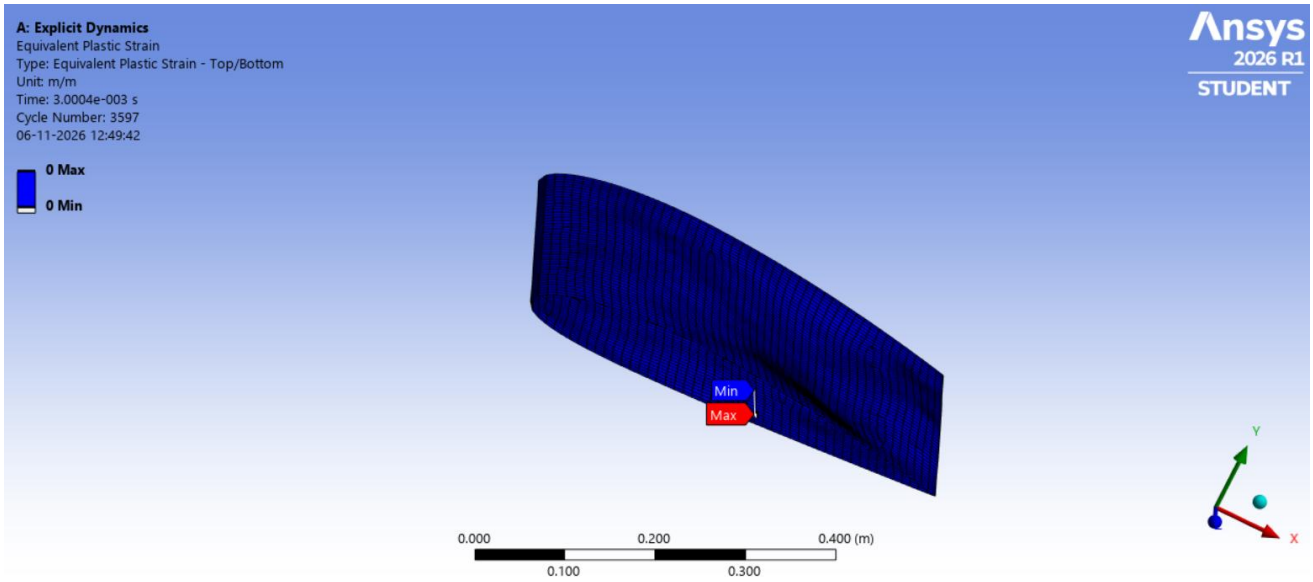


Figure 4. Equivalent plastic strain on the wing at 3.0004 ms. The displayed value is zero throughout the shell.

The 1,064.8 MPa displayed stress is an elastic-model output. The material law does not account for stress redistribution through yielding. The zero plastic strain follows from the elastic formulation and cannot establish that the physical wing would remain undamaged.

Comparison with the published impact case

Walvekar et al. [1] report two speeds for the 12.7 mm nose-radius specimen. Case II at 219.66 m/s is the closest to the present run by speed. The following comparison uses the paper's bird model description and Case II setup.

Parameter	Present model	Walvekar et al. Case II [1]
Bird geometry	Flat-ended cylinder	Cylinder with hemispherical ends, 98 mm diameter and 287 mm overall length
SPH resolution	8 mm particle-size setting	4 mm particle pitch
Impact	213.66 m/s, head-on along +X	219.66 m/s, reported 30° impact
Initial kinetic energy	39.7 kJ from solver output	43.7 kJ from nominal bird mass and speed
Target and material	NACA 2412 shell, linear elastic aluminium	Built-up leading edge with 12.7 mm nose radius. Al 2024-T3 with Johnson–Cook modelling.

The present impact speed was 2.7% lower and the initial kinetic energy was approximately 9% lower. These are differences in the input conditions. The impact angle, bird shape and target construction also affect load transfer.

In the published test and simulation, the skin tore along the rivet line and the ribs showed tearing and buckling [1, Table 8]. The present shell omits those joints and internal members. Figure 1 provides a qualitative point of comparison with the particle dispersion sequence in [1, Fig. 18]. Table 8 records component damage, so it provides no like-for-like values for the present stress and displacement.

Energy balance and conclusion

At the final frame the solver reported 27.29 kJ of kinetic energy, 11.05 kJ of internal energy and an energy-balance error of −1.54 kJ (−3.87% of initial energy). The final wing hourglass energy was 6.94 J, about 0.76% of wing internal energy. These checks describe the archived run. Sensitivity to shell mesh and particle resolution remains untested.

The current run demonstrates particle dispersion and transient elastic shell response. The published Case II provides a reference for comparing impact conditions and observed damage. Experimental validation of the present model would require matched test conditions and comparable response quantities.

Reference

[1] Walvekar, V., Thorbole, C. K., Bhonge, P., and Lankarani, H. M. (2010). Birdstrike Analysis on Leading Edge of an Aircraft Wing Using a Smooth Particle Hydrodynamics Bird Model. ASME IMECE2010, paper IMECE2010-37667. doi:10.1115/IMECE2010-37667.